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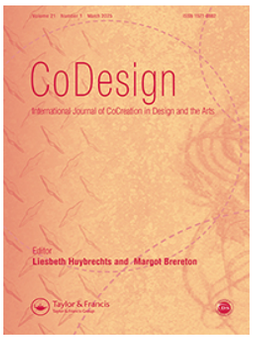
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Facilitating co-design among older adults in a digital setting: methodological challenges and opportunities

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ABSTRACT

Healthy ageing is a global priority due to a growing older population, which increases the need for preventive measures and tailored technology. In health technology development, co-design is emphasised as a valuable strategy to support a person-centred approach. Co-design, a value-driven and collaborative approach, involves end users in development processes to overcome barriers connected to capability, opportunity, and motivation. While a growing number of older adults are involved in design processes, there is a deficit of suitable methodologies for achieving active involvement. Additionally, the COVID-19 pandemic necessitated a shift to developing methodological skills and tools to facilitate co-design remotely in a digital setting. Here, we draw on experiences of conducting iterative co-design workshops with a Canadian and a Swedish cohort of older adults about technology development to support mobility, balance, and confidence in daily movement. We describe and discuss methodological and ethical challenges and opportunities to provide recommendations for conducting co-design research in a digital setting with older adults (+65 years). Our recommendations include the use of live mind mapping to facilitate participation involvement, and we address the issue of 'homework' in co-design and the importance of setting expectations.

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Co-design methods; e-health; healthy ageing; older adults; remote co-design; user participation

1. Introduction

Technology development to support older adults' (OA) ability to stay healthy, physically active, and independent is increasingly important, particularly in the face of growing ageing populations and corresponding demands on healthcare systems (Cristea et al. 2020; United Nations Department of Economic and Social Affairs UN, Population Division 2022). Through the COVID-19 pandemic, many OA have learned about opportunities in the digital landscape, where the importance of online social networks as sources of health information and social support are brought into focus (Haase et al. 2021).

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Recent research demonstrates that OA are positive towards using smart technologies, though some barriers remain – not least learning how to use digital technologies for social purposes (Harris, Blocker, and Rogers 2022). Barriers and facilitators highlighted in recent literature address capability, opportunity, and motivation, suggesting that interventions should be co-created and tailored to enhance OA digital engagement (Kebede et al. 2022). Developing technology for OA requires a broad, multidisciplinary approach to address the variety of requirements (Czaja et al. 2019; Fischer, Peine, and Östlund 2019). Methodologies that involve OA in the design process have long been recommended (Bate and Robert 2006). While such methodologies are increasingly common, more research is needed into appropriate methods for OA involvement (Cole et al. 2022; Wegener et al. 2023).

1.1. Co-design in e-health development

Participatory methods for healthcare technology design have been applied to develop, adapt, and optimise interventions to end users such as patients and healthcare providers (Bate and Robert 2006). However, a recent review shows that the methods used to involve patients in development and improvement processes usually involve questionnaires and interviews, which are typically considered low-level engagement tools (van der Voorden et al. 2023). Methods to increase patient engagement are needed, particularly in the digital setting, which can offer access and flexibility for individuals with varied abilities and needs. Addressing the complexity of healthcare and the challenge of implementing new tools requires deep anchoring amongst care providers in situ (de Plaza et al. 2023; Nusir and Rekik 2022).

Co-design is a methodology that can formally incorporate the ideas and values of end users into the development of services or interventions. Engaging end users early in the development process makes it possible for them to express their needs and aspirations (Neves and Vetere 2019) and to adapt the developed healthcare service/intervention to fit the context and users' needs, avoiding research waste (Slattery, Saeri, and Bragge 2020). Using co-design in healthcare technology development enables close collaboration between the key stakeholders, the facilitation team, and technology developers (Bate and Robert 2006), but there is a lack of literature on methodological and ethical challenges in engaging OA in remote co-design.

1.2. Conducting co-design in the digital landscape

The COVID-19 pandemic forced many researchers to conduct their data collection remotely. A study regarding online co-design with children Fails et al. (2022) showed that there were challenges and advantages associated with this digital transition, such as a pronounced power dynamic between children and adults, and increased opportunities for new ways of communicating. Based on their experiences, Fails et al. (2022) suggested that COVID-19 led to new ways of handling the digital setting associated with future co-design. When engaging OA in this digital setting, some challenges and advantages might be associated with age, technology literacy, and access to information and communication technology (Olsson, Samuelsson, and Viscovi 2019). An age-related digital gap can still be found, dependent on contributing factors such as older adults' personal attitudes,

motivations, education, income, and social support (Song, Qian, and Pickard 2021). Other influencing factors among OA may be accessibility, usability, access to a stable internet connection, and associated expenses (e.g. device, network subscription). It is important that researchers and developers are aware of this age-related digital gap, and that the use of technology can both empower and disempower OA. Non-use, lower use, or poor use of digital technology may lead to exclusion and can be detrimental for OA outcomes (Neves and Vetere 2019). Engaging OA when designing and developing digital technologies and services is therefore crucial and should be implemented early in the development process.

In co-design, processes, methods, and tools to support creative engagement need to be tailored to accommodate the use, context, and skills of the participants (Galabo and Cruickshank 2022). When in-person activities are not allowed or are difficult to accomplish, a remote digital setting enables participation, helping to ensure that essential perspectives or participants are not excluded. However, there is still a lack of knowledge on how to tailor methods, tools, and settings to support OA creative engagement in remote co-design activities in a safe and ethical way. In this paper, we draw upon research experiences where we had to adjust to a digital setting due to the COVID-19 pandemic.

This paper discusses issues that arose when conducting co-design research remotely with OA, during the COVID-19 pandemic.

Specifically, we address the 1) challenges and opportunities to consider when involving OA in remote co-design of an e-health technology tool, 2) ethical challenges of the online co-design process, and we 3) provide recommendations for conducting co-design activities with OA in digital settings.

2. Co-design methodology

Our research approach addressed three principles of human centred design, participatory design, and co-design, as described by Steen (2011). First, our design process sought to understand the needs, perspectives, and lived experiences of people who will use or be impacted by the technologies or services designed. Second, we engaged directly with people to achieve this understanding. Third, our primary role was as facilitators to help people express their needs, perspectives, and lived experiences in a way that could inform the development of new technologies and services.

2.1. Overall design

Our major technological goal was to develop a feasible and effective peer-to-peer learning intervention within the primary healthcare setting, supporting OA in improving physical function and confidence in daily movement to prevent mobility deterioration, and maintain independence and active living. The project followed the Medical Research Council framework for complex interventions (Craig et al. 2008; Moore et al. 2015) focusing on Development, Feasibility, and Evaluation (Figure 1). We involved OA (+65 years), in an iterative series of co-design workshops (WS) aiming to identify their needs and adapt and refine the intended intervention based on their input. The co-design experiences in this paper come from the project's first phase (Development), where the focus was on needs assessment, trust-building, and adapting and refining the

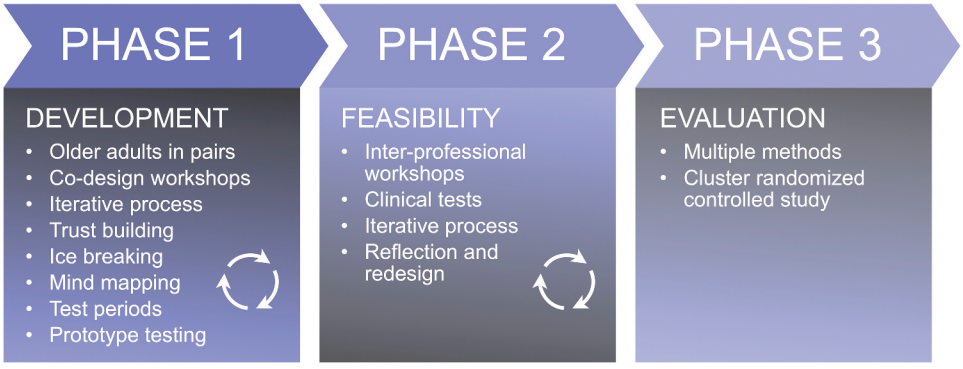


Figure 1. The data collection/co-design workshops in this paper were conducted during the first phase of the larger project.

intervention (see Figure 1). The aim of the first phase of the project was to co-design a digital tool/peer-to-peer learning intervention among older adults to regain physical functioning and confidence in daily living. The research questions were: What are the OA needs of digital tools that may enhance their physical function and confidence in daily living? How can a video-based platform be co-designed with OA to improve physical function and confidence in daily living? How can OA be involved in improving the design of such a video-based platform? In this paper, we draw upon the methodological findings and reflections from nine iterative co-design workshops conducted in 2020–2021 with two cohorts of participants. The content-related outcome will be published in another paper.

The COVID-19 pandemic hit during the planning stage of the first phase of the project, and we had to adapt to conduct the co-design workshops in a digital setting. We share our reflections from this transition and provide recommendations based on our experiences and current literature.

2.2. Participants and setting

This study was a result of a collaboration between the University of Calgary (Canada – CAN) and Linnaeus University (Sweden – SWE). Two cohorts of participants, referred to as co-designers, were recruited, one from each country. The co-designers were organised in pairs, consisting of OA who could be partners or friends, whether they lived together or not. It was essential that these pairs maintained regular contact with each other due to the peer-to-peer component of the intervention. Each cohort included six OA, totalling three pairs. The CAN co-designers (Table 1) were recruited through personal contact or posters at the university; inclusion criteria included being able to move independently with or without walking aids and being >65 years old. The CAN co-designers were physically active, without any mobility impairments, and had previous experience handling different technologies. The SWE co-designers were recruited through an ongoing, closely related study on telemonitoring devices at home (Ekstedt et al. 2023) with the additional inclusion criterion of having one or more chronic illnesses. The SWE co-designers reported chronic conditions such as heart failure or hypertension. Two of them

Table 1. Demographic information of co-designers.

	Age	Relationship
Canadian cohort		
Pair 1	78 (M), 74 (F)	Partners, living together
Pair 2	70 (M), 72 (F)	Partners, not living together
Pair 3	66 (F), 68 (F)	Friends, not living together
Swedish cohort		
Pair 4	68 (F), 68 (F)	Friends, not living together
Pair 5	77 (M), 77 (F)	Partners, living together
Pair 6	78 (M), 77 (F)	Partners, living together

had mobility impairments and needed walking aids, while three had more than one chronic condition (Table 1). Proficiency in English and Swedish was required for the CAN and SWE groups, respectively.

Although the co-design workshops were initially planned to be in-person, they were adapted to an online Zoom format due to COVID-19 safety measures. Four workshops were conducted with the CAN cohort (WS-CAN) from November 2020 to February 2021, and five workshops were conducted with the SWE cohort (WS-SWE) from February 2021 to May 2021.

The Canadian co-designers participated in all their workshops. However, illness and technological issues prevented some Swedish co-designers from participating in all workshops (see Table 2) and one couple withdrew from the study prior to the first workshop due to the lack of reliable internet access. All participants provided informed consent prior to their first workshop (WS1) in accordance with the approval from the Research Ethics Board in Canada (REB:20–0003) and the Swedish Ethical Review Authority (DNR:2020–01219).

2.3. Facilitation structure

The facilitation team consisted of one senior lecturer (project leader, SB), two doctoral students (GF, SS), and two senior professors (LK, ME). SB and GF led the Canadian workshops, while SB and SS led the Swedish workshops. The facilitation team had diverse backgrounds, including physiotherapy, nursing, health technology design, and educational psychology, bringing expertise in qualitative research methods, co-design, and technology development. However, conducting co-design workshops remotely was a new experience for all researchers.

Before the inaugural workshop, the facilitators conducted online meetings with each co-designer pair to ensure they had a functional technology setup and were familiar with

Table 2. The Canadian and Swedish cohort co-designer attendance.

	Canadian cohort (Nov 2020 - Feb 2021)				Swedish cohort (Feb 2021 - May 2021)				
	WS1 (CAN)	WS2 (CAN)	WS3 (CAN)	WS4 (CAN)	WS1 (SWE)	WS2 (SWE)	WS3 (SWE)	WS4 (SWE)	WS5 (SWE)
Pair 1	X	X	X	X					
Pair 2	X	X	X	X					
Pair 3	X	X	X	X					
Pair 4					X	X	X		X
Pair 5						X	X	X	X
Pair 6					X	X	X	X	X

Zoom usage. In accordance with general co-design principles (Czaja et al. 2019; Sanders and Stappers 2008), the first workshop focused on exploring the co-designers' needs and general ideas on technology use.

Before the workshops, the facilitation team developed a baseline activity plan. This plan was iteratively revised in response to developments during each workshop, while keeping the overall project goals in mind. This flexible approach to activity planning aligned with the principles of co-design, which focuses on enabling ongoing participation of the co-designers throughout the process (Bate and Robert 2006).

After each workshop, the facilitation team conducted a debriefing session to reflect on both the structure of the facilitation and the content of the discussions (see Figure 2 and Appendix A). These debriefing sessions played a crucial role in enhancing the iterative process and gaining insights into the co-designers' experiences. Guided by these debriefing sessions, the facilitators made refinements to the workshop structure. This included a re-evaluation of the activities to be included, how they were introduced, the integration of software tools to facilitate these activities, and the development of guidelines for effective tool utilisation. Additionally, the facilitators adjusted the workshops to ensure active engagement with the workshop topics, involving modifications to workshop objectives and the questions asked.

The discussions and interactions in both cohorts were fluid and lively indicating that the OA co-designers felt comfortable and safe. The OA co-designers built on each other's reflections and expressed that it was inspiring to listen to each other's stories and experiences.

2.4. Icebreaking and needs assessment

Providing a clear description of the project's aims and clarifying expectations while creating a safe and creative atmosphere is crucial in the co-design process (Botero and Sampsa 2013; Gheduzzi et al. 2021). The first workshop (WS1) thus commenced with an introductory presentation by the facilitators about the project, followed by a Q&A session and icebreaking activities. The CAN cohort participated in a discussion of general life enjoyment, while the SWE co-design cohort speculated on which animal they would like to be for a day.

Facilitators initiated a dialogue towards a shared understanding of key concepts such as self-care, healthy ageing, and daily physical activities. Crucially, for needs assessment,

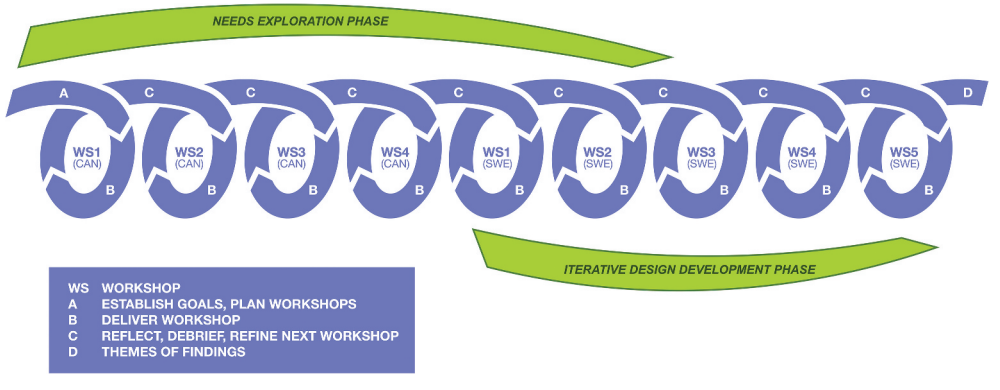


Figure 2. The iterative co-design process with the two cohorts, conducted from Nov 2020 to Nov 2021.

the facilitators encouraged co-designers to share instances from their daily lives where they felt at ease moving around, and situations that made them feel insecure or unstable. Additionally, co-designers were invited to share ideas on what could be beneficial to their mobility, balance and confidence in daily movement.

2.5. Independent tasks – 'homework'

Notably, co-designers were given tasks to be completed outside the scheduled sessions. The term 'homework' was used for this, though it connotes challenges elaborated on in the discussion. The 'homework' in preparation for the first workshop with the CAN cohort (WS1-CAN) included taking photos of those situations in their daily life where they felt confident and enjoyed moving, and those where they would like to feel more confident and safer. None of the co-designers completed the task, though they could refer to such situations throughout the workshop.

Co-designers were not pressed on why they didn't complete the 'homework', yet unprompted they noted that they had forgotten to do the task or did not have time to complete it. Indeed, some felt that they were a 'bad student'. Still, the CAN co-designers thought they could meet the aim of the 'homework' without taking any photos. For facilitators, this sparked a discussion about the implicit power relations in assigning 'homework', the potential burden of 'homework', and the need to provide resources to complete the 'homework'.

In response, facilitators actively adjusted the approach to 'homework'. For example, co-designers in the SWE cohort did not take photos before the first session but we still invited them to complete other 'homework' assignments. Everyone in the CAN cohort completed the 'Choose at least one of the selected technologies and test at home' but only some completed the 'Pick and test an app at home from the provided list'. Everyone in the SWE cohort completed 'Test the prototype and reflect', while some completed 'Find and test a digital application that supports your health'. Co-designers in both cohorts were asked to use a physical notepad for their experiences and reflections between workshops. Interestingly, no co-designers completed this, offering explanations that they forgot about it, found it time-consuming, or found it easier just to talk about their experiences.

2.6. Live mind mapping

In the briefing session after WS1-CAN, the facilitation team decided to use WS2-CAN to discuss how technology might help a person regain and improve physical function and confidence in daily movement. The decision was made to test the use of a mind mapping software Xmind (Xmind Ltd 2023) that remained visible to all co-designers throughout the WS2-CAN, to track and facilitate the discussion. One facilitator (SB) led the discussion, while the other (GF) recorded insights brought up during the discussion, developing the mind map in session. The mind map was blank at the start, while discussion started with a tentative formulation of a central topic, this was revised by the co-designers during the conversation. Ultimately, the co-designers settled on the topic 'Central Mobility Challenge: Maintaining motivation, feeling success in the technology interaction'.

As the group discussion evolved, topics were added to the mind map, sometimes as 'floating topics', letting the co-designers decide how to connect each 'floating topic' to the

mind map. Both workshop facilitators asked the co-designers to deepen, evolve, add, and adjust topics and connections, which made the live mind mapping exercise flexible and creative. To provide a visual overview of the tool and the evolution of the live mind map, two snapshots from the WS2-CAN are presented in Figures 3a (near the beginning) and 3b

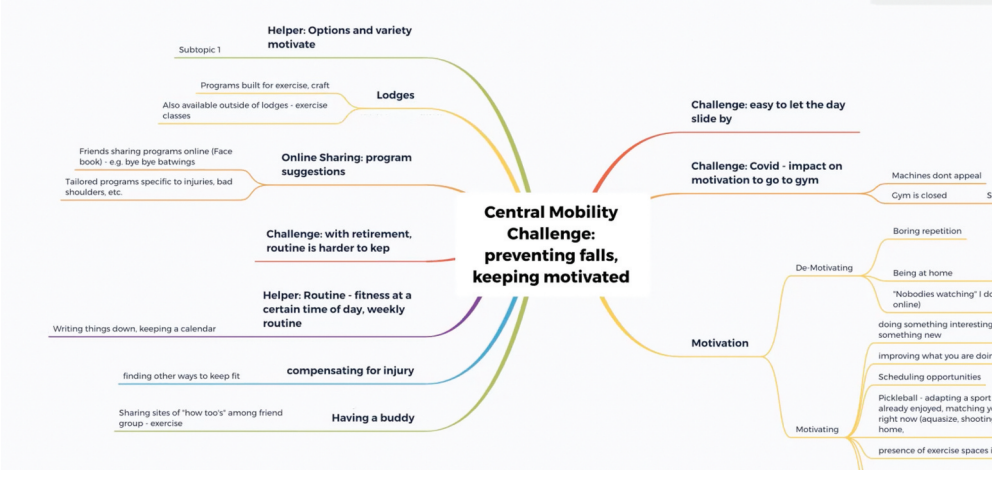


Figure 3a. Example of the live virtual mind map from early in WS2-CAN. The figure provides a screenshot from the beginning of WS2-CAN. It shows the structure of the live mind map made by one of the workshop facilitators and was visible to all co-designers.

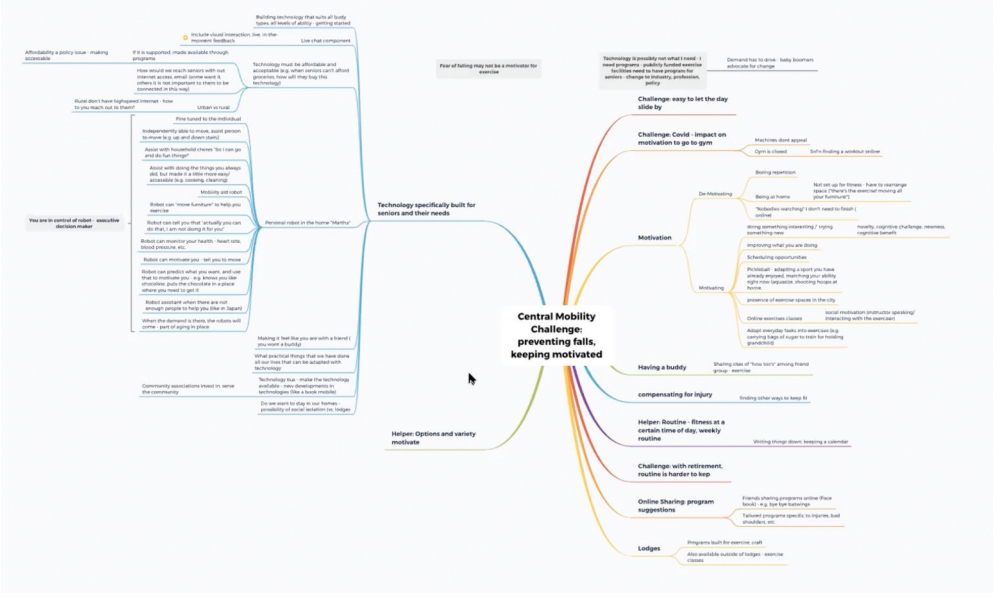


Figure 3b. Example of the live virtual mind map at the end of WS2-CAN. The figure provides a screenshot from the end of workshop WS2-CAN. It shows the structure of the live mind map made by one of the workshop facilitators (GF) and was visible to all co-designers. Floating topics were formulated before the group decided where they belonged. The central topic evolved throughout the group discussion in the workshop.

(at the end). Concluding the workshop, facilitators summarised the mind map, inviting co-designers to contribute any additions or alterations, thus providing an opportunity for instant checking (Caretta and Pérez 2019). However, it was not clear if the continuously visible live mind mapping was distracting, interrupting co-designers' trains of thought, or biasing the discussion. In the post WS4-CAN briefing, facilitators decided to explore a different approach for the Swedish cohort: periodically visible live mind mapping.

2.7. Live mind mapping – periodically visible

In WS1-SWE, the live mind map was only visible to co-designers on two occasions, right after the half-time break and at the end of the workshop. We also tested other software, a virtual Post-It note tool 'JamBoard' (Google,Jamboard 2020), allowing the use of different coloured notes, to highlight different themes. The aim of these adjustments was to learn if the periodically visible mind mapping could facilitate the discussion better than the continuously visible mind map and to compare different software tools. The findings show that when the mind map was not shown until after the half-time break, the reflections and associations between the co-designers were more fluid and creative. New and different topics were added to the discussion without co-designers being influenced by, or drawing associations to, previous topics. Indeed, co-designers noted that it was helpful for them to read and reflect on their insights based on what was written in the notes after the break and at the end. This provided an opportunity to deepen some of the themes and to clarify anything misunderstood. Co-designers also expressed a feeling of being amazed and proud at how much they had accomplished when they saw all the notes and themes.

2.8. Exploring different technology

After WS1-CAN, the facilitators curated a list of existing free apps intended to improve older adults' balance and physical activity. Facilitators provided the list during WS2-CAN and encouraged them to choose as many apps as they wanted to test on their own devices, as a basis for reflection. Only two individual co-designers tested apps from the list. Challenges with registration or downloads, lack of time, or lack of interest in the apps, radically decreased co-designers' motivation and engagement.

After WS2-CAN, facilitators decided to invite app developers to present and demonstrate different tools: Bodiometer Pro (AnimationLeader Technology Inc 2020), MyDartfish Express: Coach App (Dartfish 2019), and Move Improve (Savvy Knowledge Corporation 2019), all aimed at improving various aspects of mobility, balance, and confidence in daily movement. The tools were carefully chosen based on their broader aim, ease of use, and accessibility. Co-designers were provided with verbal and written instructions on how to access all three tools and were asked to try one or more of the tools before the next workshop. Co-designer engagement was higher in this task than in the previous 'homework', where the co-designers were asked to pick a tool from a list. In the second task, all co-designers tested at least one of the tools. This provided valuable knowledge about what tools co-designers chose to use and why, how they used them at home, and what they found valuable and not. This content related knowledge was used in the ongoing design process of a prototype and will be described in another paper.

2.9. Prototype testing and redesign

Given experiences from the workshops with the Canadian cohort, the facilitation team decided to use a video-based content delivery platform, Move Improve (Savvy Knowledge Corporation 2019), that aimed to improve mobility, balance, and confidence. The platform enables content creation. The platform had been developed and tested in other contexts by members of the facilitation team and was here used to create new content based on the co-designers' needs and wishes. The first version was developed by movement experts within the facilitation team, followed by an iterative co-creation process based on the co-designers' experiences in testing different versions. An iPad with the app installed was given to each pair in the Swedish cohort, adhering to guidelines for COVID-19 in force at the time. Co-designers were encouraged to use the app together as a pair after WS3-SWE, WS4-SWE, and WS5-SWE. Use of the app varied between pairs, but they all used it at least once. Pairs provided valuable feedback during the workshops on the content and usability of the app, allowing further development. The technology skill level within the Swedish cohort varied more than the Canadian cohort and the co-designers needed more personal support from the facilitation team between the workshops to handle updates and internet connections. Still, handing a pre-configured physical device to the co-designers resulted in more frequent use than that seen in the Canadian cohort, despite the differing skill levels.

3. Discussion

3.1. The digital setting

Involving OA in remote co-design introduces both challenges and opportunities. Leveraging a digital setting can simplify re-design efforts and enhance flexibility to accommodate co-designer's needs. Moreover, many digital platforms come equipped with built-in accessibility features such as automated closed captioning, potentially improving the involvement of older adults. With improved accessibility and increased participation opportunities for older adults, our findings confirm previous research that online co-design provides several benefits (Kennedy et al. 2021). It offers geographical and scheduling flexibility, empowering co-designers from different locations to collaborate as equals in the research process (Darley and Carroll 2022).

Darley and Carroll (2022) emphasise the importance of engaging OA with different levels of digital literacy in digital settings. Indeed, our invitation to software developers to present their apps to the Swedish cohort, with instructions, effectively enhanced co-designer engagement. Thus, if co-design facilitators can thoughtfully mitigate obstacles associated with technology, our research suggests that remote co-design can enable OA to make valuable and creative co-design contributions. However, if not fully considered, our findings show that technological difficulties can arise from multiple sources, such as access to adequate computer hardware, familiarity with video conferencing or other software, and access to a stable internet connection.

3.2. Building trust and capability

Given our findings, it seems very important to nurture the relationship between the facilitator and the remote co-designer, and to recognise that co-design is a shared

learning journey. Trust-building in co-designing with OA has been emphasised in the literature (Machado, Costa, and Mealha 2021). Carroll et al. (2021) highlight mutual learning, mutual awareness and reciprocal contribution while Hsu et al. (2024) focus on building capacity, confidence, capability and competence through a mutual learning approach. We were able to build trust through a range of activities and exercises requiring active participation – including technology aids used throughout the workshops that helped co-designers maintain their engagement and follow discussions. This is in line with Fischer et al. (2021) who highlight the significance of various learning aspects when involving OA in co-designing projects. Their findings showed that power relations, problem settings, and socio-material arrangements matter for learning outcomes, supporting the idea that facilitators should be flexible and open to incremental learning.

In line with Darley and Carroll (2022), our findings highlight the importance of preparatory measures when conducting co-design with OA remotely. Preparation demonstrates respect for co-designer participation, thus important aspects such as informing co-designers about the topic and project expectations, building rapport while maintaining good research practices, assessing technological literacy levels and support requirements, and allocating time and resources to provide support and training can facilitate trust-building and enhance capability. We found that the preparation meetings had a calming effect on both the co-designers and the facilitation team, through relationships and rapport building., Using online tools to visualise the group's achievements increased the co-designers' sense of capability and pride, aligning with findings in Falk Erhag et al. (2022).

The workshop facilitators had to repeatedly remind remote co-designers who shared the same physical room about being respectful when other people are talking. One facilitator chaired each online session to ensure that speaking time was shared roughly equally among co-designers. As highlighted (Ostrowski, Breazeal, and Park 2021) trust and mutual respect are equally important between participants and facilitators as among participants independent of the setting. Having all co-designers connected with separate devices in different locations could have improved mutual respect. However, sitting together can build capacity and reduce anxiety about technology problems, as there is a possibility to help and calm one another, and facilitate help if needed. An alternative to manage communication challenges that can arise with group workshops would be to meet with older adults individually, echoing findings in previous research (Ostrowski, Breazeal, and Park 2021). However, our findings also show that being in a group with other OA can be inspiring for OA co-designers, and that discussions between co-designers can elevate the process. Lastly, advising co-designers to manage distractions, such as phone calls, in the digital environment is important.

3.2.1. Recommendations

While the facilitator's role is to lead the workshop, technologies used to support remote co-design must reinforce an atmosphere where co-designers feel valued as true partners, alongside technology developers or researchers. We recommend effective preparation, but not tightly planning too much in advance, rather considering and clearly defining the learning goals and workshop activities to support the intended learning.

We recommend investing extra time into the introduction phase when co-designing in a digital setting, to ensure that all co-designers are comfortable in their role and that

they feel capable of handling any situational technical issues. We recommend using icebreaking activities that relate to something familiar, safe, or playful, such as talking about what makes you happy or what animal you would like to be for a day. This helps the group start talking and get to know each other and creates a relaxed atmosphere. However, it is important to continue to reinforce trust and transparency throughout the rest of the co-design process.

Our recommendation is to be wary of having several people join a session from the same room when co-designing remotely with OA, and to set expectations beforehand regarding the rules of conduct in the digital setting. As OA may have very different technology skill levels, we recommend providing everyone with the same information (including troubleshooting support) so as not to publicly embarrass anyone by asking if they need help.

3.3. Facilitating tools in the digital setting

Synchronously taking notes with an online mind mapping software tool during the co-design discussions provided a range of benefits, such as encouraging active participation and helping the co-designers return to and reflect on previous ideas, and mediated collaborative knowledge construction, with co-designers building on each other's ideas as discussed by Näykki and Järvelä (2008). This is comparable to comic boarding, a co-design technique where participants describe their thoughts and ideas while an artist presents them on paper (Moraveji et al. 2007) and is valuable in engaging older adults in co-design (Xie et al. 2012). Arguably, synchronous live mind mapping provides a feasible and accessible way of comic boarding suitable for the digital setting. Notably, facilitators found that continuously visible mind mapping sometimes disrupted or biased discussion towards certain topics, given the co-designers' focus on what was documented. In sessions where the mind mapping was not visible until the halftime break, confirming reflections and insights at that point, discussions flowed more smoothly.

The artefact resulting from synchronous documentation – the mind map itself – was also valuable to facilitators and co-designers after the workshop in which it was created. The facilitation team used the mind maps in briefing discussions between workshops, to develop a cohesive understanding of each workshop, and to help make decisions on how to proceed and adapt subsequent workshops in the unfolding iterative process. Additionally, the facilitators used the virtual mind mapping at the beginning of each workshop to summarise the previous workshop for the co-designers and get everyone on track smoothly.

Notably, remote co-designing requires flexible planning and an openness to unfolding circumstances (Darley and Carroll 2022). Facilitators must adjust and adapt the methods and the tools used to support those methods to the needs of each group of co-designers and the flow of the ongoing discussions. As we found, allowing different remote co-design methods can enhance older adults' involvement in technology development processes.

3.3.1. Recommendations

We recommend using online visualisation tools to enhance discussion and keep all co-designers on track. Using such online tools live, in a periodically visible way, can support

co-designers' reflection process without disturbing ongoing discussions. We further recommend investing time at the beginning of the process to help co-designers grow comfortable with the different remote collaboration tools and techniques.

We also encourage workshop facilitators to be familiar with different online tools, enhancing flexibility and the visualisation of discussions/activities, as it may be more difficult to capture what is going on in a remote setting. Such flexibility may help co-designers feel included even if technology glitches disrupt participation.

3.4. Co-design involvement – the issue of 'homework'

As previously discussed, building trust and creating a permissive atmosphere are significant, but when and how to involve people in co-design is also important. Power relations may influence the degree of genuine involvement of vulnerable populations in co-design (Fails et al. 2022; Hodson, Svanda, and Dadashi 2023). The importance of engaging older adults early in the co-design process and with activities based on their real-life experiences has previously been highlighted (Ghorayeb, Comber, and Gooberman-Hill 2023). To integrate real-life experiences, various independent tasks or 'homework' assignments are commonly given to co-designers to complete in their daily lives (Hsu et al. 2024). However, there is a lack of discussion regarding the implications of these tasks in terms of power dynamics and the potential pressure placed on participants.

In this project, the co-designers in both cohorts were given 'homework' assignments, though they often did not complete them. Co-designers provided multiple reasons why 'homework' was not completed, as noted above. Such challenges impacted their enthusiasm and willingness to do 'homework'. This sparked a discussion within the research team about the appropriateness of giving OA 'homework' between co-design workshops, the ethical considerations of assigning additional work outside of sessions, and the language used to describe such work. Indeed, did homework place an undue burden on OA co-designers and did the term 'homework' imply a position of authority for facilitators over the OA? Further inquiry into the reasons why OA co-designers did not complete 'homework' were beyond the scope of this research project. Potential reasons for non-completion of 'homework', such as the emotional and cognitive impacts of living through a global pandemic on motivation, were raised. The facilitation team acknowledged the prevalence of 'homework' tasks in co-design but noted limited reporting on 'homework' completion as a pattern in the literature. As such, the facilitation team could not be certain how prevalent the issue of non-completion of 'homework' was among co-designers. The benefits and burdens, terminology and structure of co-design 'homework' is an ongoing discussion within our research team and represents a potential avenue for further research and exploration.

Notably, feeling actively involved and able to contribute are important aspects for OA co-designers (Fischer, Peine, and Östlund 2019). 'Homework' is often intended to support the involvement and engagement of co-designers, engendering a sense of ownership over the co-design process and reinforcing their role as valued contributors. However, this may also create tension as the 'Homework' may have placed an additional burden on our co-designers that they did not want, were not prepared for, and for which they did not understand the value. While 'Homework' can potentially encourage

participation and engender a sense of ownership and value in the role of a contributor, it can also imply an ethical risk when co-designing with OA: 'Homework' can place an unwanted burden on OA co-designers, for whom time and energy may be a limited resource. The facilitation team also discussed whether the term 'independent task' instead of 'Homework' would have enhanced co-designer agency and released them from any sense of guilt associated with incomplete work. Finally, we discussed whether providing OA co-designers with physical materials such as disposable cameras to document places and situations where they felt confident/not confident (as in WS1-CAN), or a physical notepad to record their experiences and reflections (as in WS1-WS4-SWE), might have had an impact on the number of co-designers who completed the 'Homework'. As noted by Kramer et al. (2021), investing time and resources in co-design materials can impact the motivation of co-designers.

3.4.1. Recommendations

We recommend working with co-designers to manage expectations and identify appropriate levels of engagement and performance of tasks outside of the co-design workshops. This can ensure that the remote co-design process with OA is responsive to co-designer needs and is ethically appropriate. Indeed, engaging co-designers in the unfolding co-design process itself can be as important as their focus on a given design exercise. We further recommend designing 'homework' tasks to involve minimal logistical work, such as app registration, for a more seamless co-designer experience. It is important to focus on the larger aims of the homework, to better enable flexible co-designer participation outside of the scheduled sessions. We call on facilitators to sensitively monitor the completion of homework to build a better, and evolving, understanding of how and when to assign homework in the co-design process.

We suggest that facilitators place greater emphasis on understanding the expectations of co-designers regarding their level of commitment in the co-design process. This will allow facilitators to understand how much activity or involvement can reasonably be expected of co-designers and if there is a need to communicate the value of additional commitment. This is especially important in remote co-design, where it can be difficult to interpret body language information about a participant's level of comfort with what is being asked of them.

4. Conclusion

Remote co-design in a digital setting offers a potential means of engaging OA in co-design processes, while posing unique methodological and ethical challenges. Even though the digital setting can add additional challenges to the co-design process, the overall goal in the co-design is the same – to engage the end users and explore their lived experiences and ideas.

In this paper, we have highlighted challenges and opportunities, and provided recommendations for future co-design processes. When engaging OA in co-design and technology development, awareness of the heterogeneity regarding the use of technology, and differences in needs and wishes for the future, is important and requires a flexible and open approach in the development process. A focus on

being respectful and building capability with dignity is therefore particularly important.

Based on the findings from this research project, a summary of our methodological recommendations for conducting co-design in a digital setting with OA is presented below:

- Allocate time and effort for preparation with each of the co-designers before group sessions, this builds trust and a sense of capability, and facilitates a creative and permissive atmosphere.
- Clarify expectations and roles in the co-design process as well as proper conduct for collaboration in the digital setting, for example setting up the physical space to limit distractions, prioritising the virtual experience, and not interrupting conversations.
- Be flexible and explore different software tools to visualise the discussions and co-design activities. A periodically visible means of visualising discussions may avoid disrupting ongoing discussions.
- It's a shared journey, be open to incremental and mutual learning, do not plan too much in advance, and clarify the learning goals for each session. Briefing sessions between the workshops are essential for reflection and planning subsequent steps.
- Assess and address OA co-designers' expectations on process, 'homework' tasks, and level of engagement. Consider the terminology used to describe and emphasise the aim of the task.

Disclosure statement

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References

AnimationLeader Technology Inc, Bodiometer. 2020. https://apps.apple.com/se/app/bodiometer-pro/id1441230362.

Bate, P., and G. Robert. 2006. "Experience-Based Design: From Redesigning the System Around the Patient to Co-Designing Services with the Patient." *BMJ Quality and Safety* 15 (5): 307–310. https://doi.org/10.1136/qshc.2005.016527.

Botero, A., and H. Sampsa. 2013. "Ageing Together: Steps Towards Evolutionary Co-Design in Everyday Practices." *CoDesign* 9 (1): 37–54. https://doi.org/10.1080/15710882.2012.760608.

Caretta, M. A., and M. A. Pérez. 2019. "When Participants Do Not Agree: Member Checking and Challenges to Epistemic Authority in Participatory Research." *Field Methods* 31 (4): 359–374. https://doi.org/10.1177/1525822X19866578.

Carroll, S., K. Kobayashi, M. N. Cervantes, S. Freeman, M. Saini, and S. Tracey. 2021. "Supporting Healthy Aging Through the Scale-Up, Spread, and Sustainability of Assistive Technology Implementation: A Rapid Realist Review of Participatory Co-Design for Assistive Technology with Older Adults." *Gerontology and Geriatric Medicine* 7:233372142110232. https://doi.org/10.1177/23337214211023269.

Cole, A. C., K. Adapa, A. Khasawneh, D. R. Richardson, and L. Mazur. 2022. "Codesign Approaches Involving Older Adults in the Development of Electronic Healthcare Tools: A Systematic Review." *British Medical Journal Open* 12 (7): e058390. https://doi.org/10.1136/bmjopen-2021-058390.

Craig, P., P. Dieppe, S. Macintyre, S. Michie, I. Nazareth, and M. Petticrew. 2008. "Developing and Evaluating Complex Interventions: The New Medical Research Council Guidance." *BMJ* 337: a1655. https://doi.org/10.1136/bmj.a1655.

Cristea, M., G. G. Noja, P. Stefea, and A. L. Sala. 2020. "The Impact of Population Aging and Public Health Support on EU Labor Markets." *International Journal of Environmental Research and Public Health* 17 (4): 1439. https://doi.org/10.3390/ijerph17041439.

Czaja, S. J., W. R. Boot, N. Charness, and W. A. Rogers. 2019. *Designing for Older Adults: Principles and Creative Human Factors Approaches*. 3rd ed. Florida: CRC Press.

Darley, A., and Á. Carroll. 2022. "Conducting Co-Design with Older People in a Digital Setting: Methodological Reflections and Recommendations." *International Journal of Integrated Care*. 22 (4): 18. https://doi.org/10.5334/ijic.6546.

Dartfish, myDartfish Express: Coach. 2019. https://apps.apple.com/us/app/mydartfish-express-coach-app/id1040982427 .

de Plaza, P., M. Alejandra, L. Yadav, and A. Kitson. 2023. "Co-Designing, Measuring, and Optimizing Innovations and Solutions within Complex Adaptive Health Systems." *Frontiers in Health Services* 3:3. https://doi.org/10.3389/frhs.2023.1154614.

Ekstedt, M., E. S. Nordheim, A. Hellström, S. Strandberg, and H. Hagerman. 2023. "Patient Safety and Sense of Security When Telemonitoring Chronic Conditions at Home: The Views of Patients and Healthcare Professionals - a Qualitative Study." *BMC Health Services Research* 23 (1): 581. https://doi.org/10.1186/s12913-023-09428-1.

Fails, J. A., D. K. Ratakonda, N. Koren, S. Elsayed-Ali, E. Bonsignore, and J. Yip. 2022. "Pushing Boundaries of Co-Design by Going Online: Lessons Learned and Reflections from Three Perspectives." *International Journal of Child Computer Interaction* 33:100476. https://doi.org/10.1016/j.ijcci.2022.100476.

Falk Erhag, H., U. Lagerlöf Nilsson, T. Rydberg Sterner, and I. Skoog. 2022. "A Multidisciplinary Approach to Capability in Age and Ageing." In *International Perspectives on Aging*. Edited by J. L Powell and S. Chen. Vol. 31. Springer.

Fischer, B., B. Östlund, N. K. Dalmer, A. Rosales, A. Peine, E. Loos, L. Neven, and B. Marshall. 2021. "Co-Design As Learning: The Differences of Learning When Involving Older People in Digitalization in Four Countries." *Societies* 11 (2): 66. https://doi.org/10.3390/soc11020066.

Fischer, B., A. Peine, and B. Östlund. 2019. "The Importance of User Involvement: A Systematic Review of Involving Older Users in Technology Design." *The Gerontologist* 60 (7): e513–e523. https://doi.org/10.1093/geront/gnz163.

Galabo, R., and L. Cruickshank. 2022. "Making it Better Together: A Framework for Improving Creative Engagement Tools." *CoDesign* 18 (4): 503–525. https://doi.org/10.1080/15710882.2021.1912777.

Gheduzzi, E., N. Morelli, G. Graffigna, and C. Masella. 2021. "Facilitating Co-Production in Public Services: Empirical Evidence from a Co-Design Experience with Family Caregivers Living in a Remote and Rural Area." *Health Services Management Research* 34 (1): 21–35. https://doi.org/10.1177/0951484820971452.

Ghorayeb, A., R. Comber, and R. Gooberman-Hill. 2023. "Development of a Smart Home Interface with Older Adults: Multi-Method Co-Design Study." *JMIR Aging* 6:e44439. https://doi.org/10.2196/44439.

Google LLC. 2020. Jamboard [Mobile application software]. https://apps.apple.com/se/app/jamboard/id1143591418.

Haase, K. R., T. Cosco, L. Kervin, I. Riadi, and M. E. O'Connell. 2021. "Older Adults' Experiences with Using Technology for Socialization During the COVID-19 Pandemic: Cross-Sectional Survey Study." *JMIR Aging* 4 (2): e28010. https://doi.org/10.2196/28010.

Harris, M. T., K. A. Blocker, and W. A. Rogers. 2022. "Older Adults and Smart Technology: Facilitators and Barriers to Use." *Frontiers in Computer Science* 4. https://doi.org/10.3389/fcomp.2022.835927.

Hodson, E., A. Svanda, and N. Dadashi. 2023. "Whom Do We Include and When? Participatory Design with Vulnerable Groups." *CoDesign* 19 (4): 1–18. https://doi.org/10.1080/15710882.2022.2160464.

Hsu, L.-J., P. B. Stafford, W. Khoo, M. Swaminathan, K. J. Amon, H. Sato, K. M. Tsui, D. J. Crandall, and S. Sabanović. 2024. ""Give it Time:" Longitudinal Panels Scaffold Older Adults' Learning and Robot Co-Design." *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*. https://doi.org/10.1145/3610977.3634979.

Kebede, A. S., L. Ozolins, H. Holst, and K. Galvin. 2022. "Digital Engagement of Older Adults: Scoping Review." *Journal of Medical Internet Research* 24 (12): e40192. https://doi.org/10.2196/40192.

Kennedy, A., C. Cosgrave, J. Macdonald, K. Gunn, T. Dietrich, and S. Brumby. 2021. "Translating Co-Design from Face-to-Face to Online: An Australian Primary Producer Project Conducted During COVID-19." *International Journal of Environmental Research and Public Health* 18 (8): 4147. https://doi.org/10.3390/ijerph18084147.

Kramer, L. L., M. Blok, L. van Velsen, B. C. Mulder, and E. de Vet. 2021. "Supporting Eating Behaviour of Community-Dwelling Older Adults: Co-Design of an Embodied Conversational Agent." *Design Health (Abingdon)* 5 (1): 120–139. https://doi.org/10.1080/24735132.2021.1885592.

Machado, S., L. Vale Costa, and Ó. Mealha. 2021. "Co-Designing with Senior Citizens: A Systematic Literature Review. Human Aspects of it for the Aged Population." In *Technology Design and Acceptance: 7th International Conference, ITAP 2021, Held as Part of the 23rd HCI International Conference, HCII 2021, Virtual Event, July 24–29, 2021, Proceedings, Part I*. https://doi.org/10.1007/978-3-030-78108-8_5.

Moore, G. F., S. Audrey, M. Barker, L. Bond, C. Bonell, W. Hardeman, L. Moore, et al. 2015. "Process Evaluation of Complex Interventions: Medical Research Council Guidance." *BMJ: British Medical Journal* 350:h1258. https://doi.org/10.1136/bmj.h1258.

Moraveji, N., J. Li, J. Ding, P. O'Kelley, and S. Woolf. 2007. "Comicboarding: Using Comics As Proxies for Participatory Design with Children." *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. https://doi.org/10.1145/1240624.1240832.

Näykki, P., and S. Järvelä. 2008. "How Pictorial Knowledge Representations Mediate Collaborative Knowledge Construction in Groups." *Journal of Research on Technology in Education* 40 (3): 359–387. https://doi.org/10.1080/15391523.2008.10782512.

Neves, B. B., and F. Vetere. 2019. *Ageing and Digital Technology Designing and Evaluating Emerging Technologies for Older Adults*. Singapore: Imprint: Springer.

Nusir, M., and M. Rekik. 2022. "Systematic Review of Co-Design in Digital Health for COVID-19 Research." *Universal Access in the Information Society* 23 (2): 1–15. https://doi.org/10.1007/s10209-022-00964-x.

Olsson, T., U. Samuelsson, and D. Viscovi. 2019. "At Risk of Exclusion? Degrees of ICT Access and Literacy Among Senior Citizens." *Information, Communication & Society* 22 (1): 55–72. https://doi.org/10.1080/1369118X.2017.1355007.

Ostrowski, A. K., C. Breazeal, and H. W. Park. 2021. "Long-Term Co-Design Guidelines: Empowering Older Adults As Co-Designers of Social Robots. 2021." *Proceedings of the 30th IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*. https://doi.org/10.1109/RO-MAN50785.2021.9515559.

Sanders, E. B. N., and P. J. Stappers. 2008. "Co-Creation and the New Landscapes of Design." *CoDesign* 4 (1): 5–18. https://doi.org/10.1080/15710880701875068.

Savvy Knowledge Corporation, Move Improve. 2019. https://apps.apple.com/us/app/move-improve/id1434897217.

Slattery, P., A. K. Saeri, and P. Bragge. 2020. "Research Co-Design in Health: A Rapid Overview of Reviews." *Health Research Policy and Systems* 18 (1): 17. https://doi.org/10.1186/s12961-020-0528-9.

Song, Y., C. Qian, and S. Pickard. 2021. "Age-Related Digital Divide During the COVID-19 Pandemic in China." *International Journal of Environmental Research and Public Health* 18 (21): 11285. https://doi.org/10.3390/ijerph182111285.

Steen, M. 2011. "Tensions in Human-Centred Design." *CoDesign* 7 (1): 45–60. https://doi.org/10.1080/15710882.2011.563314.

United Nations Department of Economic and Social Affairs UN, Population Division. 2022. *World Population Prospect 2022 - Summary of Results. UN DESA/POP/2022/TR/NO.*

van der Voorden, M., W. S. Sipma, M. F. C. de Jong, A. Franx, and K. Ahaus. 2023. "The Immaturity of Patient Engagement in Value-Based Healthcare-A Systematic Review." *Frontiers in Public Health* 11:1144027. https://doi.org/10.3389/fpubh.2023.1144027.

Wegener, E. K., J. M. Bergschöld, C. Whitmore, M. Winters, and L. Kayser. 2023. "Involving Older People with Frailty or Impairment in the Design Process of Digital Health Technologies to Enable Aging in Place: Scoping Review." *JMIR Human Factors* 10:e37785. https://doi.org/10.2196/37785.

Xie, B., T. Yeh, G. Walsh, I. Watkins, and M. Huang. 2012. "Co-Designing an E-Health Tutorial for Older Adults." *Proceedings of the 2012 iConference*, Toronto, ON: Canada. https://doi.org/10.1145/2132176.2132207.

Xmind Ltd. 2023. Xmind, Ver.23.07 [Mobile application software]. https://xmind.app/blog/xmind-2023-new-version-updates/.